Design Docs

This document contains design docs for the Imbalance team. Each design doc is saved as a subpage below.

260331: Imbalance at Risk (IaR) — Concept and Methodology

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This document is split across two pages due to length. See the subpage for Sections 5–7, Goals, Non-Goals, Assumptions, Open Questions, and Plan.

Issue

Electricity market participants accumulate net imbalance positions in every Market Time Unit (MTU) as a result of forecast errors in production, consumption, or both. These positions are settled against the TSO-published imbalance price, which can be highly volatile and extreme during system stress. Despite this, most market participants manage their imbalance exposure informally or reactively. There is no standard product that quantifies portfolio-level imbalance settlement cost risk in a statistically rigorous way, comparable to how financial institutions manage Value at Risk. Optimeering's imbalance price forecast distributions — covering the full quantile range from P01 to P99 — make it possible to build exactly such a product. The Imbalance at Risk (IaR) framework defined in this document provides the conceptual and mathematical foundation for doing so.

Background

Imbalance settlement in European electricity markets

Each balance responsible party (BRP) holds a net imbalance position in each MTU:

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net imbalance(t) = (actual − contracted) generation(t)
                 + (actual − contracted) consumption(t)
```

where consumption is signed negative by convention. If a BRP produces more than contracted (long) or consumes less than contracted (short), the net surplus is settled against the imbalance price published by the TSO.

Settlement conventions: Nordic (NO/SE/FI/DK) uses single-price settlement (Regulating Power Price); Germany uses single-price rebalancing energy; Belgium/Netherlands use two-price settlement in some periods. This framework initially assumes **one-price settlement**, applicable to Nordic and German markets.

Optimeering's forecast products

Optimeering produces probabilistic forecasts of the imbalance settlement price for each MTU, up to 36-48 hours ahead: expected value plus quantiles P01, P05, P10, P25, P50, P75, P90, P95, P99. These define the **marginal predictive distribution** at each MTU individually, but do not directly encode the **joint distribution** across MTUs — how prices at different MTUs co-move over time.

The problem with naive per-MTU risk assessment

Two crucial effects are ignored by per-MTU risk assessment:

1. **Temporal autocorrelation of imbalance prices.** Prices are not independent across MTUs. System tightness tends to persist: high up-regulation prices in one MTU predict tightness in nearby MTUs. Ignoring this leads to incorrect period-level aggregation.
2. **Position-price dependence.** A BRP's position and the prevailing imbalance price are often correlated. For example, a wind-dominated portfolio tends to be short precisely during low-wind periods, which are also periods of higher system prices — so the portfolio's worst imbalance positions coincide with the most expensive settlement prices. Ignoring this leads to systematic underestimation of the true IaR.

A statistically rigorous IaR requires modelling the **joint distribution** of prices and positions across the full time horizon of interest.

Solution / Design

1. Mathematical definition of IaR

Symbol	Meaning
t	Index for a Market Time Unit (MTU)
T = {t1, ..., tn}	The set of MTUs over the horizon of interest

lambda_t	Imbalance settlement price at MTU t (EUR/MWh)
p_t	Day-ahead market (DAM) price for delivery at MTU t (EUR/MWh)
s_t	Imbalance price spread at MTU t: s_t = lambda_t - p_t
q_t	Net imbalance position at MTU t (MWh); positive = long, negative = short
pi_t	Imbalance settlement P&L at MTU t (EUR)

MTU P&L (one-price settlement): `pi_t = q_t * lambda_t` [gross], `pi_t_spread = q_t * (lambda_t - p_t)` [spread]

Portfolio P&L over horizon T: `PI_T = sum_t(q_t * lambda_t)` [gross], `PI_T_spread = sum_t(q_t * (lambda_t - p_t))` [spread]

Two IaR variants: `IaR_alpha(T) = -Q_alpha(PI_T)` [Gross IaR], `SIaR_alpha(T) = -Q_alpha(PI_T_spread)` [Spread IaR]

Gross IaR = absolute exposure to settlement costs; relevant for cash flow and liquidity. **Spread IaR** = incremental exposure vs. DAM benchmark; the more economically meaningful metric for performance assessment. Both are computed, reported, and subject to independent limit structures.

IaR_alpha(T) is the maximum expected loss over horizon T exceeded with probability at most alpha. IaR is a positive number representing a cost (loss). Per-MTU IaR variants exist as building blocks but do not correctly aggregate to period IaR without the joint distribution.

Relationship between Gross and Spread IaR when DAM price is known (intraday): For Tier 1 (deterministic positions), D = sum_t(q_hat_t * p_t) is a known constant and SIaR_alpha(T) = IaR_alpha(T) - D. For a long position in a positive-price market, D > 0 and Spread IaR < Gross IaR — the DAM revenue (partially) offsets the imbalance settlement exposure. For stochastic positions (Tiers 2/3) no simple relationship holds.

When DAM price is uncertain (day-ahead planning horizon): Before the DAM clears, p_t is a random variable and Spread IaR requires the joint forecast distribution of (lambda_t, p_t) including their correlation (typically positive, since both reflect supply-demand tightness). Volue produces DAM price forecasts with quantile distributions. Optimeering's pre-DAM imbalance forecasts will soon incorporate Volue's DAM expected value forecasts as features, giving automatic mutual consistency between the two inputs — an important design advantage for the day-ahead Spread IaR extension. The primary use case remains intraday with known DAM prices.

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2. The joint distribution problem

IaR_alpha(T) requires the joint distribution of all q_t and lambda_t across all MTUs in T — a 2n-dimensional distribution. For a 24-hour horizon with 15-minute MTUs (both Nordic and German markets use 15-minute settlement), this is 192-dimensional (96 MTUs x 2 variables). For a 36-hour day-ahead horizon: 288-dimensional. This high dimensionality motivates the copula-based approach over explicit covariance parameterisation.

By **Sklar's theorem**: F(lambda_t1, q_t1, ...) = C(F_lambda_t1(lambda_t1), F_q_t1(q_t1), ...) — separating the problem into:

- **Component A: Marginal price distributions** — Optimeering's quantile forecasts per MTU
- **Component B: Marginal position distributions** — customer-provided (three tiers, Section 3)
- **Component C: Dependence structure** — price autocorrelation, position autocorrelation, price-position cross-dependence; calibrated from historical data

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3. Three tiers of position data

Tier 1 — Deterministic point forecast: q_t = q_hat_t. All uncertainty from price. IaR understated if position has material uncertainty. Baseline for most customers; still valuable for correct period IaR via price autocorrelation.

Tier 2 — Independent marginal distributions: q_t ~ F_q_t independently per MTU. Captures within-MTU uncertainty; misses temporal autocorrelation. A persistent forecast error (e.g. systematic wind over-forecasting during a high-pressure front) would produce correlated position errors — not captured here. Achievable with reasonable effort.

Tier 3 — Full joint position distribution: (q_t1, ..., q_tn) ~ F_q with temporal autocorrelation encoded. From stochastic simulation of the customer's generation/consumption model. The correct representation. Optimeering may offer a side product to assist customers in reaching Tier 3.

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4. Period IaR vs per-MTU IaR aggregation

Naive aggregation (incorrect): Summing per-MTU IaRs assumes worst-case losses occur simultaneously — overly conservative and structurally wrong (no well-defined probability level for the period).

Correct period IaR: IaR_alpha(T) = -Q_alpha(sum_t q_t * lambda_t) from the full joint simulation.

Diversification ratio: IaR_alpha(T) / sum_t IaR_alpha(t), in (0, 1]. Approaches 1 when prices are highly positively autocorrelated (Nordic regulation events); materially lower in normal conditions. A key product metric — shows how limit headroom varies with market regime.

Marginal IaR: MIaR(t|T) = partial IaR_alpha(T) / partial q_t. Identifies which MTUs drive the period risk exposure.

Sections 5–7, Goals, Non-Goals, Assumptions, Open Questions, Plan

5. Monte Carlo scenario generation

Recommended methodology: historical copula with forecast marginals (filtered historical simulation) — calibrates dependence structure from historical data while using current Optimeering forecast distributions as marginals.

Step 1 — Calibrate price copula from historical data. Compute rank-transformed uniform scores u_{t,s} = F_hat_lambda_t(lambda_{t,s}) for each historical day s. The matrix of uniforms encodes historical rank-correlation without assuming a parametric marginal. Fit a Gaussian copula via the normal scores transform, or use block-bootstrap resampling as a non-parametric alternative.

Step 2 — Fit forecast marginals. For each MTU t, fit a parametric distribution to the 9 Optimeering quantile points. Candidate families: Normal (likely too symmetric), skew-normal/skew-t, Johnson's SU (flexible four-parameter, good tail fit), or non-parametric empirical interpolation.

Step 3 — Generate price scenarios. Draw N uniform vectors from the copula. Transform to prices: lambda_t^(s) = F_lambda_t_inv(u_t^(s)). Produces N price paths with marginals matching current forecasts and autocorrelation calibrated to history.

Step 4 — Generate position scenarios (tier-dependent). Tier 1: q_t^(s) = q_hat_t (deterministic). Tier 2: q_t^(s) = F_q_t_inv(v_t^(s)), v_t uniform independent. Tier 3: customer-provided stochastic scenarios via same copula-and-marginal approach.

Step 5 — Future: Joint price-position scenarios. Price and position drawn jointly using a cross-copula calibrated from historical co-movements, addressing the price-position dependence problem. See Open Questions.

Step 6 — Compute IaR. For each scenario s: PI_T^(s) = sum_t q_t^(s) * lambda_t^(s). IaR = negative empirical alpha-quantile over N scenarios. N >= 1000 for alpha=0.05; N >= 5000 for alpha=0.01.

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6. Limit structures

All limit types apply independently to both Gross IaR and Spread IaR, independently monitored and alerted.

Limit types supported: Per-MTU IaR limit (IaR_alpha(t) <= L for all t); Period IaR limit (IaR_alpha(T) <= L for defined T); Per-MTU IaR in session (all t in remainder of trading day); Rolling window IaR (any rolling m-MTU window); Marginal IaR limit (MIaR(t|T) <= L, limits per-MTU contribution to period IaR).

Horizons configurable: remainder of current trading day, next trading day, user-defined MTU range, rolling m-MTU window. Limits set per IaR variant (Gross/Spread), per confidence level alpha, per price area, optionally per sub-portfolio.

Spread IaR limits are the more natural choice for economic risk management intraday (p_t known). Gross IaR limits remain relevant for cash-flow and collateral management.

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7. Real-time monitoring and alerting

As each MTU is delivered it moves from forecast to actual — the remaining horizon shrinks and IaR recalculates. Update cadence: 15 minutes (Nordic and German markets).

Alert conditions: Hard breach (IaR > L, immediate notification); soft warning (IaR > 0.80 * L, advance warning); trend alert (rate of change of IaR is high, breach likely within k MTUs).

Delivery: webhooks (REST/JSON) to customer systems; API endpoints for polling; standalone monitoring UI (Optimeering-hosted, no customer integration required).

Update triggers: new Optimeering forecast publication; customer position update; MTU passage; explicit customer request.

Goals

- Provide a statistically rigorous, VaR-equivalent measure of imbalance settlement cost risk for BRPs in target markets
- Correctly capture cross-MTU price autocorrelation so that period-level IaR comes from a well-defined joint distribution, not a naive sum of per-MTU VaRs
- Correctly capture price-position joint distribution including cross-dependence
- Support all three tiers of customer position data quality with a clear upgrade path
- Enable configurable limit structures for both Gross and Spread IaR, with real-time alerting
- Deliver both API/webhook integration and standalone monitoring UI
- Launch in Nordic markets with architecture extensible to Germany and other EU markets

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Non-Goals

- UK imbalance markets (GB cash-out mechanism requires separate treatment)
- Two-price settlement markets in initial version (Belgium etc.) — flagged as future extension

- Financial hedging recommendations
- Position forecasting (ingests forecasts; does not generate them)
- Intraday trading optimisation
- Credit or counterparty risk

Assumptions

- Optimeering's quantile forecasts are well-calibrated (P05 exceeded ~5% of the time historically)
- Target markets have at least 2 years of MTU-level historical imbalance price data for copula calibration
- One-price settlement applies in initial target markets
- Customer position data can be ingested with latency less than 1 MTU duration (15 minutes)

Open Questions

1. Price-position joint distribution (high priority for v2): How to model Corr(q_t, lambda_t)? Options: (a) empirical calibration from customer historical data; (b) structural model based on portfolio type; (c) default independence with disclosed conservative bias. Materially affects IaR accuracy for portfolios with structural correlation.
2. Copula family selection: Gaussian copula underweights joint tail events. Student-t (low degrees of freedom) or Clayton copula may better capture extreme stress scenarios. Validate against historical data per market.
3. Marginal distribution fitting from quantiles: With 9 quantile points, tails beyond P01/P99 are extrapolated. Sensitivity analysis comparing Normal, skew-t, and Johnson's SU needed in POC.
4. Number of scenarios: N=1000 is fast but noisy at alpha=0.01; N=10,000 more stable but higher latency. Consider quasi-Monte Carlo or stratified sampling.
5. Nordic two-price settlement periods: Some NO areas apply different prices for long vs. short positions during reserve activations. Must be reviewed against current TSO rules.
6. Real-time position data API: Format, latency, authentication, update frequency. Does the customer push updates or does the product poll?
7. Realised IaR tracking: Should the system track actual imbalance costs alongside forecast IaR and report ex-post exceedances? Useful for limit reporting and model validation.
8. Multi-area portfolios: Some BRPs operate across multiple Nordic price areas. Requires modelling cross-area price correlation. Initial version may scope to single-area.
9. Day-ahead Spread IaR (before DAM clears): When p_t is uncertain, Spread IaR requires the joint forecast distribution of (lambda_t, p_t). Optimeering's pre-DAM imbalance forecasts will soon incorporate Volue's

DAM expected value forecasts as features, providing automatic consistency between inputs. Natural extension but adds significant complexity; initial version assumes intraday with known DAM prices.

Plan

Phase 1 — Scoping and concept development (current): Define IaR methodology precisely (this document); develop illustrative mock-ups (IaR timeline view, limit dashboard, alert interface); identify representative customer scenarios (Nordic hydro, Nordic wind); validate methodology against historical data for 1-2 Nordic price areas.

Phase 2 — Presentation and stakeholder alignment: Produce 25-30 minute customer-facing presentation covering the problem, methodology, product, value proposition, and roadmap. Audience: operational planners, risk managers, traders, portfolio managers. Include worked examples and illustrative IaR calculations from real or synthetic data.

Phase 3 — Proof of Concept: Build end-to-end pipeline (price forecast ingestion to scenario generation to IaR computation); implement at least one limit type with alerting; build basic visualisation (IaR over remaining day, per-MTU breakdown); validate against at least one historical stress event (large Nordic imbalance price spike); document API specification for position ingestion and alert delivery.